

Energy Analytics - Forecasting Report

May 2026

1 Introduction

This is the final report for the forecasting competition assignment in the Energy Analytics course. The competition entailed prediction of log-demand of electricity in England and Wales in eight separate days throughout the duration of the course. On each day, the parameters of a normal distribution, μ and σ , were submitted and scored according to a log-likelihood scoring function. The evaluation metric therefore penalized both point-estimate inaccuracy and poor uncertainty calibration, requiring a balance of maximizing the sharpness of predictions while also managing the uncertainty of the predictions.

This report outlines the approach used to generate forecasts, including the mistakes made along the way, as well as the justification for the modeling decisions.

2 Data Exploration

The dataset covers the period from January 2020 to the present (at the time of making the predictions) and comprises three components. First, daily total electricity demand for England and Wales, published by National Grid, measured in MWh. Second, historical weather observations from Meteoblue at 6-hourly resolution (midnight, 6am, noon, 6pm) for London, Bristol and Leeds and daily sunshine duration in minutes for each city. Third, weather forecasts on the same basis as the historical observations, covering both temperature and sunshine duration, available from 13 January 2025 to 23 March 2025 and 10 January 2026 onward. These forecasts are used in place of observed weather data when constructing features for the live competition forecasts.

Simple visualisation of time series log demand highlights the seasonal but also weekly pattern it follows. These patterns seem to be to large extent predictable as changes in demand due to key dates such as Christmas holidays are clearly discernible, showing a very similar pattern each year. This motivates inclusion of date-specific variables to be used in predictions such as day-of-the-week, month, but also indicator that the day is a holiday. Additionally, the data exhibit a structural decline in demand, suggesting that long-term time trend should be considered when forecasting. Another important observation is that the demand pattern during COVID lockdown, despite exhibiting a somewhat steeper decline in the spring of 2020, does not diverge significantly from the behaviour of demand in post-COVID period. This contributed to the decision of including the COVID data in the training sample, to maximise the length of the training set and preserve the model's ability to learn from low-demand regimes that may recur during holiday periods or periods of unusually mild weather.

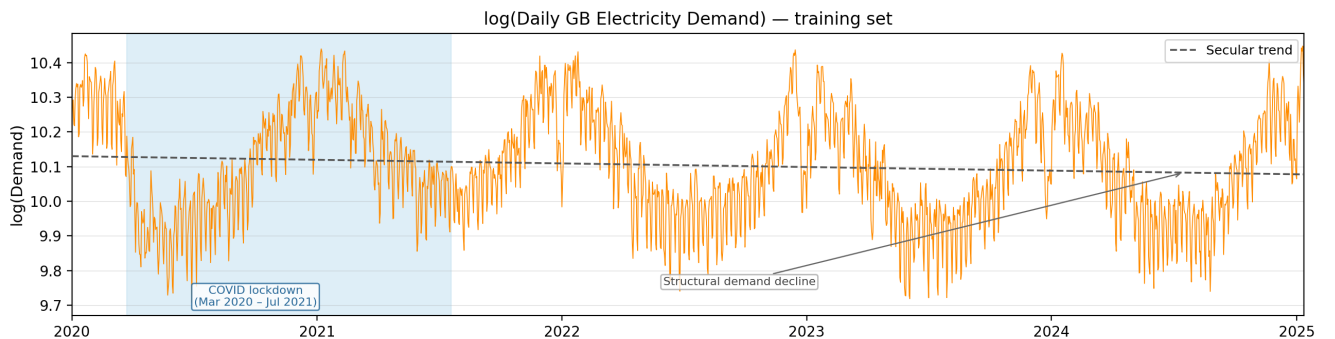


Figure 1: Log-demand time series

Plot of log-demand against the average daily temperature (taken from the four data-points for each day) highlights a non-linear relationship. Below approximately 15 degrees, the demand exhibits negative linear relationship with temperature, meaning that as temperatures fall, demand rises. However, above 15 degrees, demand becomes far less sensitive to temperature, with only a mild positive slope which seems to start at around 20 degrees and above. This asymmetric shape is rather intuitive as during cold days, heating of houses increases demand substantially, while during very warm days, air-conditioning is a driver of demand. This directly motivates use of industry standard indicators of Heating Degree Days and Cooling Degree Days as temperature features rather than raw temperatures, since a single linear term would average across two structurally distinct situations and lose explanatory power in both.

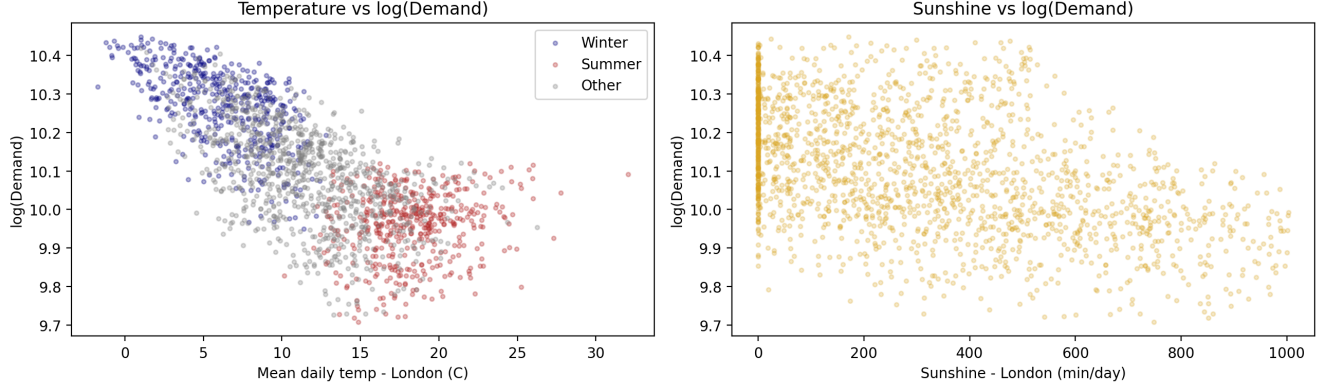


Figure 2: Temperature and sunshine vs. log-demand

Figure 3 plots the autocorrelation function (ACF) and partial autocorrelation function (PACF) of log-demand. The ACF shows strong autocorrelation persisting across all displayed lags with clear spikes at multiples of 7, reflecting the weekly demand cycle. The PACF indicates that after controlling for intermediate lags, the dominant partial correlations occur at lags 1 and 7, directly motivating the inclusion of lags among the demand features in Section 4. However, due to the forecast setting where we are predicting tomorrow, with demand figures only up to yesterday, the most recent available data point is lag 2.

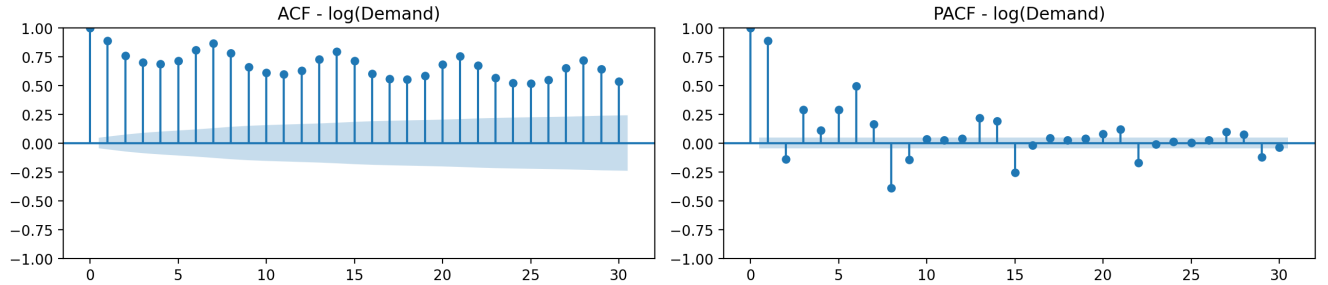


Figure 3: Autocorrelation analysis

3 Validation and Evaluation Framework

In order to estimate the expected competition performance of a model, before using it to generate the prediction, it was important to set up a proper validation and evaluation framework. The metric used to assess the performance of each model was chosen to be the the log scoring function used in the competition to exactly mimic the competition setting. Each created model had the goal of outputting the parameters of a normal distribution, μ and σ which would then be scored as follows.

$$\text{Score}(x) = -\frac{1}{2} \ln(2\pi) - \ln(\sigma) - \frac{(x - \mu)^2}{2\sigma^2}$$

where x was the actual log-demand for that day. When validating performance over a period of days, the average score was used as the metric. As a reference point, a naive climatological forecast reporting the historical mean and standard deviation of log-demand regardless of any features achieves a mean log score of approximately 0.41, providing a zero-skill baseline against which all models can be compared.

The most important thing when validating and evaluating a model that makes predictions using temporal data is not to train a model on data that would not be available during the competition. This means that the model cannot be trained on data from the future, and the log-demand data could only be used for dates up to 2 days prior to the prediction date. In order to achieve this, only log-demand data from more than two days ago was included in the features, and whenever a prediction was made during validation or evaluation, only data-points from the past were used. This approach is referred to as the expanding-window approach in this report. For each forecast date, the model is fitted from scratch on all available training data up to the information barrier. Hence, for each date, the training set grows by one observation, mimicking competition conditions where more training data will be available each day. By following this procedure, overfitting is also easier to detect since the disparity between the validation performance and training performance can be exposed and addressed appropriately. Crucially, while demand data are subject to the two-step barrier, day-ahead weather forecast data for the target date is available at forecast time and is used directly as feature inputs, reflecting the actual information set available during the competition.

When comparing different models during validation there are multiple other sources of bias that might result in overestimating the predictive power of the model. One is the use of actual temperature and sunshine data during training and validation of the models, and then using forecasted temperatures and sunshine duration for competition predictions. There is additional uncertainty and error present in these weather predictions that could degrade predictive performance. Another source of bias in the estimate of the score is selection bias. When tweaking parameters of a model or comparing different models on a validation set, the act of selecting the one with the best results biases the score upwards. This happens because random noise in the validation data introduces stochasticity into the evaluation. As a result, the theoretically superior model may not always give the highest score due to chance.

To counteract both of these effects, only dates where forecasts were available were used as validation and testing dates. That way, the predictive performance could be validated under the same conditions as in the competition. That eliminates the bias from using forecasts instead of actual weather data. To eliminate the selection bias from model selection using validation on the validation set, a separate test set was used to get an unbiased estimate of the score. It turns out there are two periods in the dataset with forecasts available. One between January 13, 2025, to March 20, 2025. This period was utilized as the validation set for model tuning and selection. The second period runs from January 10, 2026, through to the current date, and this period served as the completely independent testing set to evaluate final model performance under competition conditions. By only evaluating the final model once on the test set, an unbiased estimate of the competition performance could be achieved.

In addition to these adjustments, the test data was further excluded from the exploratory data analysis to prevent any visual or statistical data leakage from influencing the feature engineering part of the assignment. Furthermore, the calibration of σ in the prediction (explained further in section Sigma Calibration), and hyperparameter selection of the machine learning models used was also only performed on the validation dataset.

One limitation of the validation set is worth acknowledging. The January–March 2025 period is a winter and early spring regime characterised by high heating demand, low sunshine and relatively high baseline demand levels. The competition, by contrast, runs in May, when heating demand is near zero, sunshine is substantial and baseline demand is at a seasonal trough. The validation set is therefore not fully representative of competition conditions, and sigma calibrated on it may not perfectly reflect the uncertainty structure of May forecasts. This limitation reinforced the decision to use a conservative estimate of σ rather than mean-optimal multiplier, providing a degree of robustness against the possibility that the validation period understates forecast uncertainty in the competition regime.

4 Feature Engineering

The goal of the feature engineering was to represent known physical and behavioral drivers of electricity demand rather than maximize in sample fit by data mining. Therefore, as alluded to in the data exploration section, only

patterns which have a clear rational explanation were used to guide feature design. In total, 60 features were constructed out of the provided data.

To capture the asymmetric temperature-demand relationship through the mechanics that actually cause this pattern, Heating Degree Days (HDD) and Cooling Degree Days (CDD) were used. For each of the three cities, HDD is defined as $\max(0, 15.5 - \text{temp}_{\text{mean}})$ and CDD as $\max(0, \text{temp}_{\text{mean}} - 22.0)$ where the thresholds were chosen to reflect the approximate kink points visible in the scatter plot. These piecewise-linear transformations capture the heating and cooling regimes separately, allowing the model to assign different slopes to these situations. Next, diurnal spread ($\text{temp}_{\text{max}} - \text{temp}_{\text{min}}$) is included as an additional feature per city, capturing variation in the temperature profile across the day that is not reflected in the daily mean but is still predictive of demand. A large diurnal range implies a cold overnight period driving morning heating demand even on days where the afternoon mean temperature appears moderate.

While HDD and CDD capture the level of heating and cooling demand, they do not account for the lagged effect of temperatures on household behaviour. HDD_{lag1} encodes the previous day’s heating degree day, which can strongly influence current demand: if temperatures cooled significantly the previous evening, households may not have fully heated up overnight, or may begin heating only in the morning. However, the response to heating need varies between weekdays and weekends, motivating the interaction term $\text{HDD} \times \text{weekend}$, which distinguishes between demand responses dominated by commercial and industrial heating during the week and residential heating at the weekend.

During feature construction, including HDD, HDD_{lag1} and ΔHDD simultaneously was initially attempted, but this created a perfectly collinear triplet since $\Delta\text{HDD} = \text{HDD} - \text{HDD}_{\text{lag1}}$ by construction, causing numerical instability in the OLS fit. ΔHDD was consequently dropped, with HDD and HDD_{lag1} together conveying the same information without the singularity.

Additionally, length of the day in hours is derived from the date and latitude to serve as a proxy for structural lighting demand, which is independent of temperature. In winter, demand is elevated partly because darkness persists for up to 16 hours regardless of how cold it is and day length captures this effect more smoothly across the spring and autumn transition periods than month dummies alone, which impose discrete changes at calendar boundaries. Next, sunshine duration per city captures day-to-day variation in natural light within a specific season that day length alone cannot reflect. Additionally, sunshine also captures the reduction in demand caused by solar panels which are usually installed directly in the distribution network, and thus reduce the net load that must be served by the transmission network. However, since the model operates in log space, all coefficients are interpreted as multiplicative effects on demand. Thus, if sunshine reduces demand by a roughly fixed absolute amount through lighting substitution and distributed solar generation, it should translate to a larger proportional reduction, and therefore a larger coefficient, when baseline demand is low in summer than when it is high in winter. For this reason, interaction terms between sunshine and a low-HDD regime indicator for each city were created to capture this asymmetry. The low-HDD regime is defined as a binary flag equal to one when the London HDD falls below a threshold of 4.

Day-of-week dummies and month dummies are also introduced to capture the weekly and annual cycles in demand visible in Figure 2. Beyond standard calendar controls, holiday dummies are specified granularly rather than pooled into a single indicator, reflecting the substantially different demand profiles across holiday types. Christmas and New Year are separated from generic bank holidays, and Easter is decomposed into five separate indicators covering Good Friday through Easter Tuesday. Adjacent-day effects are also included since demand tends to wind down the day before a bank holiday as businesses reduce operations, and may recover only gradually the day after if people take additional days off work, motivating separate day-before-holiday and day-after-holiday indicators. Finally, a school holiday indicator captures the systematic difference in weekday demand between term-time and holidays.

Demand lags and rolling means are also included to capture persistence and seasonal structure that is not fully absorbed by the other features. Under the two-step information barrier, lag_2 is used as the closest observable demand lag. Next, lag_7 addresses the weekly serial correlation highlighted in Figure 3. For the annual seasonal anchor, lag_{364} is preferred over lag_{365} because 364 is exactly 52 weeks, preserving day-of-week alignment and thus avoiding the day-of-the-week noise that a 365-day lag would introduce. To complement this, $\text{lag}_{365_{\text{roll}7}}$, a 7-day average around the same calendar date in the previous year, provides a smoother seasonal baseline that is less sensitive to idiosyncratic demand spikes on a single day. Additionally, roll7-mean and roll28-mean, computed as rolling means of lag-2 demand, provide short- and medium-horizon estimates of the current level around which demand fluctuates.

Lastly, three additional structural features are added. Linear year trend is expressed as days elapsed since 1 January 2020 scaled by 365.25 to capture the demand decline shown in Figure 1. A COVID lockdown dummy covers the period from 23 March 2020 to 19 July 2021, isolating the structurally different demand regime. Finally, a temperature anomaly feature measures the deviation of the current London mean temperature from its long-run day-of-year climatological average, isolating the surprise component of unusual weather events. An unexpectedly cold April day will have a moderate HDD but a large negative anomaly, and it is the unexpectedness of the weather that drives additional demand as the grid responds to conditions outside the normal seasonal pattern. Two days with identical HDD values but different seasonal contexts therefore carry different demand implications, and the anomaly feature captures the residual effect that HDD alone cannot isolate.

5 OLS Regression

Given the rich set of engineered features OLS is the natural starting point for several reasons. Since demand is modelled in log space, the coefficients are interpreted multiplicatively, which aligns well with the intuition that demand drivers such as temperature and day of week act proportionally rather than in fixed absolute terms. Additionally, OLS has an exact analytical solution for both the mean and variance of the prediction distribution, aligning well with the goal of predicting the distribution of demand and the log-score evaluation criterion. The existence of closed-form solutions also enables very fast refitting under the expanding window, which is a notable benefit given that the test contains over 100 days, meaning the OLS has to be refitted over 100 times across the test period.

Using the 60 engineered features, the OLS fitted on the full training and validation set achieves an in-sample R-squared of 0.952, RMSE of approximately 3.6% and MAE of approximately 2.9%, indicating that the selected features collectively explain the large majority of variation in daily demand. Almost all major drivers are statistically significant at conventional levels (5% or lower).

The residual distribution is approximately symmetric and bell-shaped, broadly supporting the normality assumption that underpins the OLS predictive distribution formula. However, the residual autocorrelation function reveals remaining structure at lags 1 and 7, suggesting that the model does not fully capture the serial dependence in demand. This limitation motivates the stacked corrections introduced in Section 7.

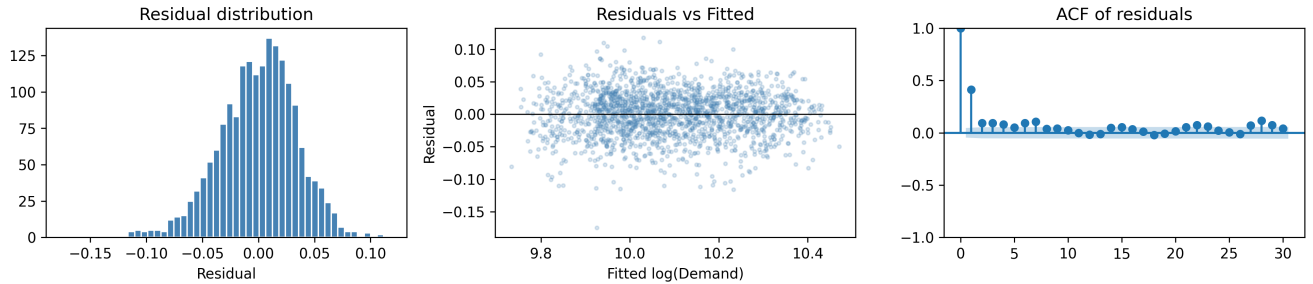


Figure 4: Residuals diagnostics

While in-sample metrics are informative, the goal is to maximize out-of-sample performance. Thus, as highlighted in Section 3, using the expanding window prediction procedure on test set with forecast weather data as inputs rather than observed temperatures and sunshine, the OLS model achieves an out-of-sample mean log score of 1.682, RMSE of 4.47% and R^2 of 0.949. This represents a substantial improvement over the climatological baseline of 0.41, which is a naive forecast that reports a normal distribution fitted to the historical mean and standard deviation of log-demand, ignoring all features entirely. The model therefore captures the large majority of forecastable variation in demand, with the remaining error attributable to residual serial dependence, nonlinear feature interactions, and the inherent unpredictability of day-to-day demand fluctuations. These figures serve as the benchmark against which the full forecasting stack in Section 7 is evaluated.

Table 1: Selected OLS coefficient estimates

Feature	Coef.	Std. Err.	t	p
<i>Temperature</i>				
HDD_london	0.0095	0.001	6.37	0.000
CDD_london	0.0136	0.003	4.18	0.000
temp_anomaly	0.0067	0.002	4.41	0.000
<i>Day of week (reference: Monday)</i>				
dow_Sat	-0.1382	0.005	-25.93	0.000
dow_Sun	-0.1473	0.005	-26.99	0.000
<i>Demand lags</i>				
lag_2	0.2135	0.022	9.53	0.000
lag_7	-0.0242	0.024	-1.01	0.313
roll7_mean	0.1701	0.042	4.04	0.000
<i>Holidays</i>				
is_holiday_christmas	-0.1167	0.011	-10.91	0.000
is_holiday_new_year	-0.0808	0.013	-6.29	0.000
is_holiday_generic_bh	-0.1191	0.009	-12.81	0.000
is_easter_tuesday	-0.0141	0.016	-0.85	0.394
<i>Structural</i>				
year_trend	-0.0035	0.001	-4.23	0.000

Table 1 shows selected key OLS coefficient estimates grouped by feature category. The temperature features carry the expected signs, as each additional HDD unit in London increases log-demand by approximately 0.95%, reflecting heating requirements, while each additional CDD unit increases it by 1.36%. The positive and statistically significant coefficient on temperature anomaly suggests that cooling demand dominates the anomaly effect, as sudden heatwaves may drive air conditioning demand beyond what CDD alone captures. However, clean interpretation of this coefficient is limited for two reasons. First, in the neutral temperature band between 15.5 and 22 degrees where both HDD and CDD are exactly zero, the anomaly feature is the only variable capturing temperature variation, meaning its coefficient may be absorbing multiple distinct effects simultaneously. Second, in a 60-feature regression there is a risk of multicollinearity absorption where the coefficient may partly reflect correlations with other features rather than a clean direct effect. Saturday and Sunday reduce demand by 13.8% and 14.7% respectively relative to Monday with very high precision. Interestingly, lag_7 is statistically insignificant, likely a consequence of multicollinearity as its effects are probably absorbed by $roll7_{mean}$ and other lag features in the model. The holiday effects reveal meaningful heterogeneity, with the impact on demand varying across holidays. Finally, the $year_trend$ coefficient implies a structural demand decline of approximately 0.35% per year, consistent with Figure 1.

6 Sigma Calibration

One important aspect during the competition was calibrating which value of σ to use for the predicted normal distribution $\log(demand) \sim N(\mu, \sigma^2)$. The basis for quantifying the uncertainty comes from the OLS regression model. For each OLS prediction t , the prediction error variance is computed as the following.

$$\sigma_t^2 = \widehat{MSE} \cdot (1 + \mathbf{x}_t^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_t)$$

The value of σ_t quantifies how uncertain the model is of its prediction and in the context of the the competition, a higher value of σ_t means that the model has a lower maximum potential score if the prediction is exactly right, but faces a less severe penalty if the prediction is off. The problem with this approach is that it does not capture the uncertainty of the weather predictions that are used as features to make the prediction. During training, these values are certain measurements, but during the competition they are forecasts.

In order to calibrate for this uncertainty, the validation and evaluation framework explained earlier in the report

was utilized. Using the validation dataset, different multiplicative values on the σ_t were tried to find the best average score on the validation set.

$$\max_m \left(\frac{1}{N} \sum_{i=1}^N \left(-\frac{1}{2} \ln(2\pi) - \ln(m \cdot \sigma) - \frac{(x - \mu)^2}{2(m \cdot \sigma)^2} \right) \right)$$

The result of running this procedure with the base OLS model is shown below in figure 5.

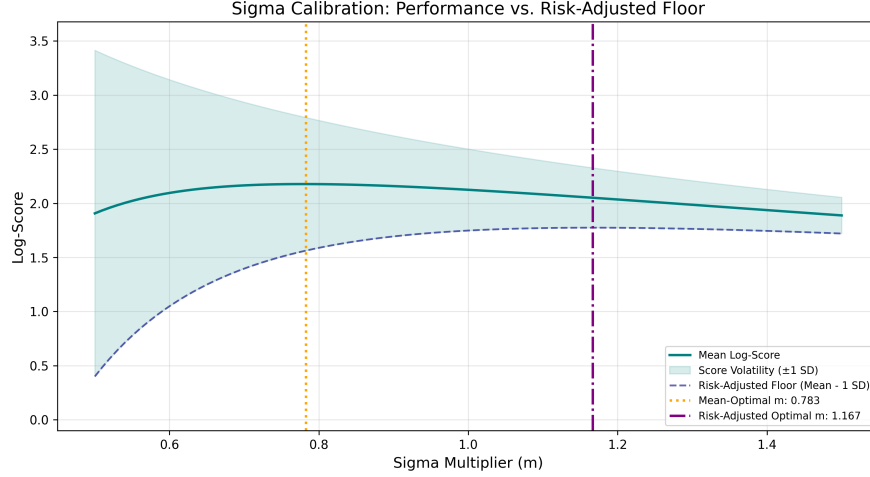


Figure 5: Calibration of σ for OLS model

In figure 5 above, the result of the sigma calibration procedure can be seen. The optimal value for m was found to be 0.783. Interestingly, this seems to show that the prediction error variance from the OLS model was conservative, even though it did not account for uncertainty in the weather forecasts. This shows that the right balance for maximizing the expected log-score does not necessarily coincide with the standard prediction error variance from the OLS model, which explains why m was not greater than 1.

However, the scoring mechanism used to evaluate the prediction distribution is asymmetric. The penalty for underestimating the variance, being overconfident and missing the target, scales aggressively compared to the penalty for overestimating the variance. For lower values of m , it can be seen that the standard deviation of the average log-score was high. Many of the scores were negative as a result of the asymmetric scoring mechanism. Since there were only eight rounds of the competition, the high variance introduces significant sampling risk. In the long term, scaling down σ by a factor of 0.783 would theoretically yield the highest score, however across such a short horizon the risk of receiving a large negative score is not negligible and would be difficult to recover from. Therefore, rather than optimizing for the mean score, a risk-adjusted floor, defined as the mean minus one standard deviation, was optimized instead. This approach sacrifices a small amount of average performance to protect the worst-case scenario. The resulting value of m was 1.167.

As the competition progressed, machine learning techniques were used to improve the point estimate by predicting the residuals of the OLS model as described in Section 7. This improved the accuracy of the predictions. With higher accuracy predictions, lowering the value of σ would be beneficial since it would result in higher scores. However, machine learning algorithms like XGBoost only optimize for and output a point estimate and this decrease in prediction error is not automatically reflected in the OLS-derived value of σ . Therefore the σ calibration procedure was run again as the model was improved. Using the final model architecture for this competition, the sigma calibration is shown below in figure 6.

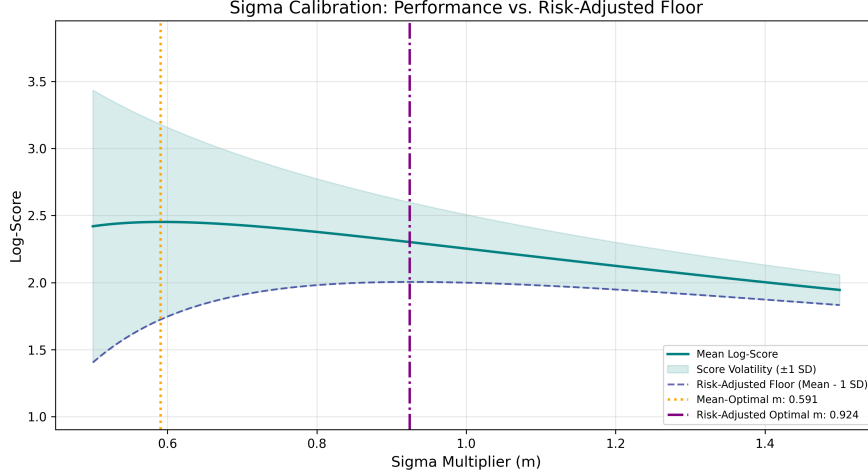


Figure 6: Calibration of σ for final model

It can be seen in figure 6 that improving the accuracy of the predictions both increased the average log-scores achieved, as well as reducing the risk-adjusted optimal m as expected. In the end, $m = 0.924$ was the value used for generating predictions.

This finding naturally follows the discussion above. Since the OLS-derived sigma has no knowledge of XGBoost and AR corrections applied on top of it, the OLS sigma becomes relatively too wide compared to the true residual uncertainty of the full stack. The calibration procedure therefore finds a lower optimal multiplier to compensate for this by reducing optimal m from 1.167 for OLS alone to 0.924 for the full stack.

7 Full Stack Model

Due to the strong performance of the OLS with 60 carefully engineered features, it is taken as a baseline forecast, with additive mean correction performed on this forecast to account for limitations of the OLS. First, regression cannot capture nonlinear feature interactions. Second, the residual ACF shows autocorrelation at lags 1 and 7 as highlighted in Figure 4.

To capture nonlinearities and interactions not represented in the OLS specification, an XGBoost model is fitted on the OLS residuals. For each forecast date, a 3-year sliding window of 1095 days of OLS residuals is used as the training target, with the same feature matrix as inputs with the exception of the year trend. Diagnostic testing showed that including the year trend caused systematic overcorrection. This is likely because tree-based models cannot extrapolate beyond the range of values seen during training, so by the time forecasts are made in 2026, the year trend value falls outside the range of the 3-year sliding window, causing XGBoost to produce a biased correction at the boundary. The model is trained with 200 trees, maximum depth of 4 and learning rate of 0.05, with prediction being added directly to the OLS mean as: $\mu_{\text{corrected}} = \mu_{\text{OLS}} + \hat{\epsilon}_{\text{XGB}}$. Here, sliding rather than expanding window is used deliberately since the objective of the XGBoost is to correct for more recent structural patterns in residuals on top of already solid model and using expanding window would dilute the recent signals with older, less relevant datapoints. Thus, XGBoost correction improves point forecast accuracy without requiring any changes to the sigma calibration since it adjusts only the mean value.

To eliminate the short-term serial correlation, which remained present even after XGBoost correction, an autoregressive model using lags 2 and 7 as regressors, denoted as AR(2,7) model was fitted on the last 90 days of the OLS + XGB residuals. The lag_2 was used since lag_1 is unavailable when forecasting. A 90-day rolling window is used rather than the full history, reflecting the assumption that recent residual dynamics are more informative than those from earlier periods, particularly around unusual weather events. The AR adjustment is added to the corrected mean: $\mu_{\text{final}} = \mu_{\text{corrected}} + \hat{\epsilon}_{\text{AR}}$

To illustrate the typical magnitude of each correction, Table 2 shows the forecast decomposition for representative

competition days. In both cases the corrections are modest relative to the point forecast, confirming that OLS captures the dominant drivers of demand and the stacked corrections serve as refinements rather than material revisions.

Date	OLS (MWh)	XGB Δ (MWh)	AR Δ (MWh)	Final (MWh)	95% PI (MWh)
Thu 14 May 2026	22,811	-107	+77	22,781	[21,129 – 24,562]
Thu 21 May 2026	21,498	-55	+226	21,669	[20,123 – 23,334]

Table 2: Forecast decomposition for two representative competition days

As shown in the Table 2, the full forecasting stack achieves an out-of-sample mean log score of 1.931, RMSE of 3.57% and R^2 of 0.950, compared to 1.682, 4.47% and 0.949 for OLS alone, which is an improvement of 0.249 log-score units and 0.90 percentage points in RMSE. The OLS-first architecture is preferred over a pure machine learning approach for a fundamental reason: OLS provides a tractable closed-form predictive distribution, whereas a standalone XGBoost model does not naturally yield a calibrated uncertainty estimate without additional methods such as quantile regression. By restricting XGBoost and AR to additive corrections on the mean, the sigma calibration developed in Section 6 remains valid and the full distributional forecast is preserved intact.

Model	MLS	RMSE (%)	OOS R^2
Climatological baseline	0.41	–	–
OLS only	1.682	4.47	0.949
OLS + XGB + AR (full stack)	1.931	3.57	0.950

Table 3: Out-of-sample forecast performance across model specifications

Figure 7 illustrates the out-of-sample probabilistic forecast over the 2026 test period. The upper panel shows the predictive mean alongside 68% and 95% prediction intervals, with the actual log-demand series overlaid in black. Analysis of log scores highlights that out of 129 days in the test set, the model achieved positive score on all but 5 days, and even during these days, the score never decreased below -1, suggesting overall robustness against large errors. The log scores obtained using forecast versus observed temperatures track each other closely throughout, with almost identical mean log scores of 1.93 and 1.94 respectively, confirming that the weather model introduces only minimal additional forecast error. This is consistent with the weather forecast being an approximately unbiased predictor of realised weather: on average it neither systematically overestimates nor underestimates temperature or sunshine, so using forecast weather inputs in place of observed values does not materially degrade the demand forecast.

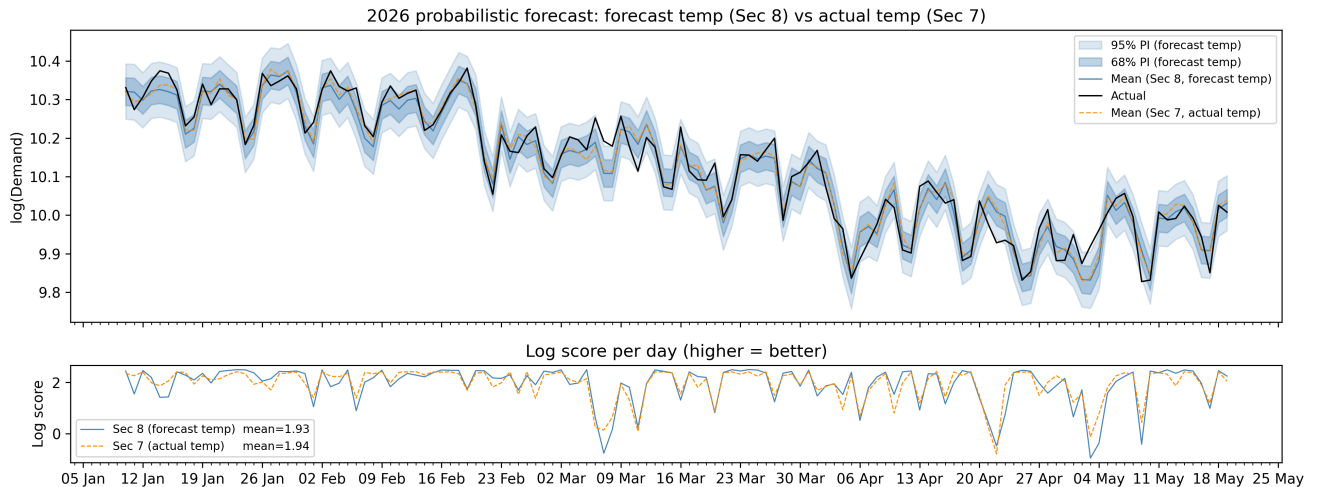


Figure 7: Test performance analysis

8 Competition Rounds

During each day of the competition, a value for μ and σ was submitted. The predictions together with the log-scores for each day is shown below in figure 8. Since the negative scores were capped at 0 in the competition, the resulting total score during the competition was 15.179.

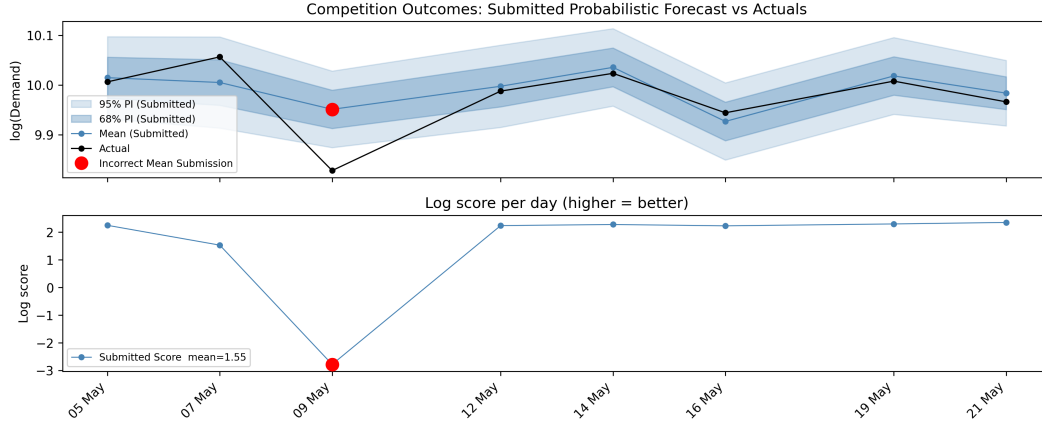


Figure 8: Competition results

Throughout the duration of the competition, the approach was altered and refined to produce better forecasts. For the first two prediction days, the basic OLS model along with the sigma adjustment method from section 6 was used. In day three, the XGBoost and Auto-regressive models were introduced and showed promise when evaluated using the evaluation framework. However, a bug was introduced into the code regarding sunshine duration data. The forecast's sunshine hours are given in hours, while the actual data is given in minutes. When adjusting the model and introducing machine learning techniques, the training and final prediction code was also altered. In the final prediction code, no adjustment was made to transform the number of predicted sunshine hours into minutes, while this was done during training. Since more sunshine generally reduced demand from the transmission network, underestimating the amount of sunshine by a factor of 60 made the model give an inflated prediction of electricity demand. As can be seen in figure 8, highlighted by the red marker, this led to a significant point estimate error for a particularly sunny day that was severely penalized. When recreating the scenario of that prediction, and using the correct features, the prediction was more closely aligned with the true value as shown in figure 9.

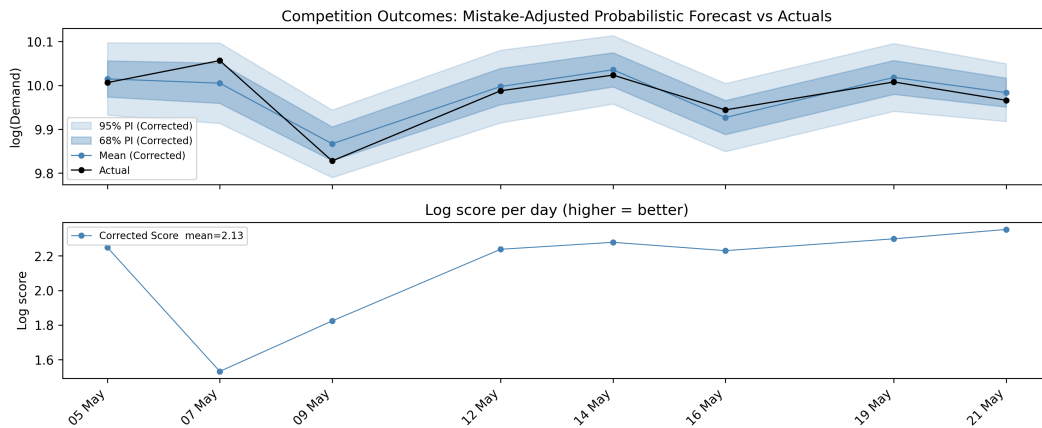


Figure 9: Competition results with correct features

In the scenario provided in figure 9, the total score would have been 17.003. This error was quickly identified and fixed before round 4 and did therefore not affect any subsequent predictions.

The same model, with the correct features, was used to generate all predictions until round 6. As it was surprising that the prediction was higher than expected given the low amount of predicted sunshine in round 7, it was found out that the XGBoost model made a large adjustment. In order to understand why, a SHAP (SHapley Additive exPlanations) analysis was conducted on the specific feature vector used for that day’s prediction. It works by evaluating what happens to the prediction when features are added or removed, considering all possible combinations of features. Using this approach, it was found that the large correction was mainly due to the year trend feature. However, because tree-based models like XGBoost cannot extrapolate continuous linear trends out-of-sample, as explained further in section 7, this strictly increasing time counter created issues for the XGBoost model. It was therefore removed from the features set only for the XGBoost and Auto-regressive models when predicting the residuals. That approach was used for the remainder of the competition.

Table 4 provides a round-by-round summary of the submitted parameters, actual log-demand, achieved log score and model specification used for each of the eight competition rounds.

Table 4: Competition rounds summary

Round	Target Date	μ	σ	$\log(y)$	Score	Model
1	05 May	10.0148	0.0412	10.0063	2.248	OLS
2	07 May	10.0051	0.0458	10.0566	1.533	OLS
3	09 May	9.9511	0.0385	9.8278	-2.787*	OLS + XGB + AR
4	12 May	9.9975	0.0414	9.9877	2.238	OLS + XGB + AR
5	14 May	10.0356	0.0390	10.0235	2.278	OLS + XGB + AR
6	16 May	9.9268	0.0388	9.9441	2.231	OLS + XGB + AR
7	19 May	10.0183	0.0386	10.0078	2.298	OLS + XGB + AR [†]
8	21 May	9.9836	0.0329	9.9661	2.352	OLS + XGB + AR [†]
Total score (negative scores capped at 0)					15.179	

* Sunshine units bug; score capped at 0 in competition.

† Year trend removed from XGBoost and AR feature set from round 7 onward.

9 Final Recommended Method

The final recommended forecasting procedure operates as follows. Daily demand and 6-hourly weather data are obtained from the dataset, with forecast temperatures and sunshine hours used for the target date and the day before that. Features are constructed as described in Section 4. OLS is refit on an expanding window basis with the training set capped at target date - 2 days, producing the baseline predictive mean μ_{OLS} . An XGBoost model is then trained on a 3-year sliding window of OLS residuals using the hyperparameters stated in Section 7, and an AR(2,7) model is fitted on the last 90 days of full-stack residuals. The final point forecast is $\mu_{final} = \mu_{OLS} + \hat{\epsilon}_{XGB} + \hat{\epsilon}_{AR}$.

The reported standard deviation is $\sigma_{calibrated} = m \times \sigma_{OLS}$, where σ_{OLS} is derived from the full predictive distribution formula in Section 6 and $m = 0.924$ is the conservative multiplier calibrated on the January-March 2025 validation set as described in Section 6. Both μ_{final} and $\sigma_{calibrated}$ are expressed in log units and submitted directly as the parameters of the normal predictive distribution over log-demand.